

GROUP PROJECT GUIDELINES

Dynamic Asset Allocation Competition

1. Project overview

The objective of this project is to design, test, and present a systematic **asset allocation investment strategy** across a predefined universe of ETFs. Each team must build a rule-based investment system capable of determining portfolio weights over time under realistic portfolio construction constraints.

Each team will compete by **submitting weekly portfolio weights** for the following four assets:

- **iShares MSCI ACWI ETF — ACWI**
- **iShares Core U.S. Aggregate Bond ETF — AGG**
- **SPDR Gold Shares — GLD**
- **Invesco Short Term Treasury ETF — BSV**

The live performance evaluation period will run from **June 1 to June 29**. During this period, the professor will track the realized performance of each team's strategy based on the portfolio weights submitted by the team.

The final project presentations will take place on **June 30**. Each team will have a maximum of **15 minutes** to present.

2. Team organization & final presentation

- The project will be carried out in **teams of 4 students**.
- Each team will develop **one single asset allocation system**.
- The work and the presentation are collective: **all members must be able to explain and defend the strategy, methodology, and results**.
- Each team must send the list of team members by **April 24, 24:00 (local time)** via email to imunarriz@faculty.ie.edu.
- From **April through May**, teams will have time to design, test, refine, and validate their investment system, as well as prepare the final group presentation.

- Submit the project materials by **June 29** to imunarriz@faculty.ie.edu.
- On **June 30**, all team members must participate in the presentation and oral defense.
- The presentation should cover the following points:

A. Strategy objective and financial rationale

B. Portfolio construction methodology

C. Backtesting and validation

D. Live implementation

- The final ranking of teams based on realized live performance will be revealed by the professor on **June 30**, after all presentations.

3. Investment universe and portfolio rules

3.1 Eligible assets

Teams must allocate capital exclusively across the following four assets, identified by their names and corresponding Yahoo Finance tickers:

- **iShares MSCI ACWI ETF — ACWI**
- **iShares Core U.S. Aggregate Bond ETF — AGG**
- **SPDR Gold Shares — GLD**
- **Invesco Short Term Treasury ETF — BSV**

No other assets are allowed.

3.2 Portfolio construction constraints

Each submitted portfolio must satisfy all of the following rules:

- The portfolio must always add up to **100%**.
- Each individual asset weight must be between **0% and 100%**.
- Portfolios must be **long-only**. Short positions are not allowed.
- Portfolio weights must be explicitly provided for all four assets.
- **These weights must be created using the model and cannot be discretionary.**

3.3 Rebalancing rule

From one week to the next, the portfolio may only be rebalanced by a maximum of **25 percentage points in total allocation change**.

In other words, teams must adjust their portfolio gradually and cannot make changes beyond the allowed rebalance limit from one submitted portfolio to the next.

3.4 Weekly submission rules

- Teams must send their updated portfolio weights **every Friday** to imunarriz@faculty.ie.edu **from May 29 to June 26**.
- The weights received on Friday will be the ones applied during the **following week**.
- The submitted allocation will remain **constant throughout that week**.
- If a team does **not** submit a new portfolio on a given Friday or if it doesn't meet any of the specified rules, the professor will assume that the applicable portfolio is **the most recent portfolio previously submitted**.

3.5 Required format for weekly portfolio submission

To facilitate processing and performance monitoring, the weekly portfolio must be submitted as a **CSV file attachment**.

The file must contain exactly **one data row** and the following header:

`week, team_id, acwi, agg, gld, bsv`

`YYYY-MM-DD, TeamXX, w1, w2, w3, w4`

Example:

`week, team_id, acwi, agg, gld, bsv`

`2026-06-06, Team03, 25, 35, 20, 20`

Submission formatting rules:

- The file must be attached in **CSV format**.
- The file name must follow this convention:
`TeamXX_YYYY-MM-DD.csv`
- Weights must be expressed as **numeric percentages without the % sign**.
- Decimals are allowed, using a **dot** as decimal separator.
- The order of columns must not be changed.
- Each weekly submission must contain only **one portfolio**, corresponding to the allocation to be implemented the following week.

3.6 Email subject for weekly submission

The subject line of the email must follow this format:

`Algorithmic Trading Project | Team XX | Portfolio for Week
YYYY-MM-DD`

Example:

`Algorithmic Trading Project | Team 03 | Portfolio for Week
2026-06-06`

4. Performance monitoring during June

Throughout June, the professor will provide teams with **weekly updates every Thursday** regarding the evolution of the performance of their portfolios.

However:

- the **composition of other teams' portfolios will not be disclosed**,
- only performance evolution will be shared,
- and the final identities of the best-performing teams will be revealed on **June 30**.

This is intended to preserve fairness and avoid imitation between teams during the live evaluation period.

5. Evaluation Criteria

The grading of the final project will follow the structure below:

- **Portfolio final performance June 29: 40%**
- **Financial logic and model complexity: 20%**
- **Backtesting, robustness, and validation: 20%**
- **Presentation quality and clarity of defense: 20%**